

Life Is Learned — Handoff 004.6

Six-Stage Lesson Architecture + Illustration Standard

Baseline: feature/catalog-001-foundation at 11742da7aa896ffb4ea77c5f2117d49fd93b5254

Create branch: feature/handoff-004-6-six-stage-lessons

Reviewer: Clyde

Status: Ready for implementation

Do not merge automatically. Stop for reviewer approval.

1. Purpose

The current app has a sound lesson player, narration, practice loop, idea cards, and authoring/timing gates, but the first five-book batch exposed two quality problems:

1. Four content screens move too quickly from concept to takeaway.
2. Artwork can technically satisfy the package contract while still feeling generic, repetitive, or decorative rather than instructional.

Handoff 004.6 establishes a stronger **six-screen authored lesson standard**, a **semantic progress indicator**, a richer **image-per-screen requirement**, and a narrow rule for when narration-follow auto-scroll should activate.

This patch must remain backward-compatible with previously imported books.

2. Six-screen authored lesson standard

Every newly authored/re-authored idea uses exactly six content screens:

Screen	Stage	PageKind	Voice	Purpose
1	Hook	.intro	guide	Create curiosity, tension, or relevance.
2	Explanation	.explanation	guide	Explain the idea clearly and accurately.
3	Story	.story	storyteller	Establish a concrete situation/example.

Screen	Stage	PageKind	Voice	Purpose
4	Story	.story	storyteller	Show the turning point, consequence, or realization.
5	Practical Application	.application	guide	Translate the idea into something usable.
6	Takeaway	.takeaway	guide	Distill the idea into the collectible bottom line.

The two Story screens are one semantic stage spread across two physical screens. After screen 6, the existing two-question practice flow begins.

Required model change

Extend:

```
enum PageKind: String, Codable, Sendable {
    case intro
    case story
    case explanation
    case application
    case takeaway
}
```

Do **not** change `formatVersion` solely for this additive enum case.

Existing stored packages that do not use `.application` must remain readable.

3. Semantic progress indicator

For lessons matching the new canonical six-screen pattern, replace the six equal capsules with:

Hook		Explanation		Story		Prac. App.		Takeaway
-		-		-----		-		-

Use width weights:

1 : 1 : 2 : 1 : 1

Visible labels:

- Hook
- Explanation
- Story
- Prac. App.
- Takeaway

These are progress identifiers, not navigation buttons.

Story segment behavior

- Screen 3 fills the first half of the wide Story segment.
- Screen 4 fills the full Story segment.
- Completed stages remain filled.
- Future stages remain inactive.

Accessibility states should preserve physical progress:

- Screen 1 of 6 · Hook
- Screen 2 of 6 · Explanation
- Screen 3 of 6 · Story 1 of 2
- Screen 4 of 6 · Story 2 of 2
- Screen 5 of 6 · Practical Application
- Screen 6 of 6 · Takeaway

At large Dynamic Type sizes, compact or adapt labels rather than breaking layout.

Legacy fallback

If a lesson does not exactly match the canonical six-page stage sequence, keep the existing per-page capsule indicator.

Do not force-migrate old books merely to obtain the new progress display.

4. Screen length and scrolling

At normal/default reading size:

- Hook should fit in one screen.
- Explanation should fit in one screen.
- Practical Application should fit in one screen.
- Takeaway should fit in one screen.
- Only Story screens are intentionally allowed to extend below the visible reader area.

This is an authoring/presentation rule, not permission to clip content.

Keep the reader scroll-safe so smaller devices, localization, and accessibility text sizes can still expose everything.

Narration-follow auto-scroll

The existing narration-follow implementation in `NarrationText.swift` already follows the spoken line in the outer `ScrollView`.

Change the call path so auto-follow is enabled by default only for `.story` pages.

Conceptually:

```
NarrationText(  
    ...,  
    followNarration: session.page.kind == .story  
)
```

Internally:

```
following = isPlaying && followNarration
```

Preserve the current safety behavior:

- do not fight user dragging/tracking/deceleration,
- respect Reduce Motion,
- do not auto-follow while VoiceOver controls the experience,
- leave prior lines visible for context.

Story narration should begin at the top and follow downward only when the spoken line approaches/leaves the visible area.

Non-story pages should not visibly drift during narration at normal size.

5. Writing budget

Six screens create better pacing, not longer lessons.

The existing hard gate remains:

300 seconds total reference experience

including narration, transitions, quiz choices, feedback, and answer allowance.

Do not speed up playback to rescue an overlong lesson.

Drafting guidance:

Screen	Suggested body prose
Hook	~25–40 words
Explanation	~40–55 words
Story A	~60–85 words

Screen	Suggested body prose
Story B	~60–85 words
Practical Application	~35–50 words
Takeaway	~18–30 words

Aim for roughly **3.5–4.5 minutes** when the material supports it. Shorter is welcome. Five minutes is a ceiling, not a target.

Stage responsibilities

Hook - Create a question, tension, or relevance. - No filler repetition.

Explanation - Build the mental model. - Distinguish author claim, evidence, uncertainty, and our application where appropriate.

Story A - Establish person, situation, and problem.

Story B - Show decision, consequence, realization, or shift. - Must advance the story rather than repeat Story A.

Practical Application - Transfer the principle to a new situation. - Give the learner something usable.

Takeaway - BLUF. - Strong enough to become the collectible idea-card front. - No generic encouragement.

6. Illustration standard

Every one of the six authored screens must have artwork.

6 content screens = 6 image references

A package cover remains separate.

Image purpose by stage

Hook - Establish problem, tension, curiosity, or setting.

Explanation - Clarify the mental model. - Conceptual/editorial is acceptable if it communicates something useful.

Story A - Show the concrete situation and recurring character/environment.

Story B - Continue the same story at a visibly different moment. - Usually one of the strongest images in the idea.

Practical Application - Show the principle being applied in a generalized/new context.

Takeaway - More symbolic/iconic is acceptable. - Must remain specific to the idea.

Reuse rules

For newly authored/re-authored lessons:

- Every page must provide `imageID`.
- Every image page must provide `imageDescription`.
- The six screens must not reuse the same `imageID`.
- The authoring gate must also reject identical resolved image bytes reused across two pages of the same idea under different IDs.
- Recurring characters, visual motifs, rooms, wardrobe, and palette are encouraged.
- Reusing the same frame is not.

Across different ideas in the same book, exact byte-for-byte image reuse should appear as an editorial warning in the authoring report rather than automatically failing the collection.

Art quality rule

Each image should answer at least one:

1. What is happening?
2. What changed?
3. What should the learner remember?

Pure decoration does not meet the editorial standard.

Avoid generated text, pseudo-book-cover typography, watermarks, and unnecessary logos inside lesson art.

7. Takeaway artwork

`ReaderView` currently renders `TakeawayCard` instead of the normal `LessonIllustration` + `NarrationText` stack.

Update `TakeawayCard` so screen-6 artwork is visible.

The visual should remain subordinate to the collectible takeaway text, but the sixth screen must not become the one image-less stage.

Keep the existing reveal interaction unless layout requires a small adjustment.

Earned Idea Card snapshot behavior from Handoff 004 remains unchanged.

8. Authoring-gate changes

Do **not** make runtime storage validation reject legacy books because they do not follow the new six-screen pattern.

The stricter rules belong in the authoring/release gate for newly authored/re-authored material.

Update `--authoring-gate` to require:

1. exactly six content pages,
2. stage sequence:

```
intro
explanation
story
story
application
takeaway
```

3. voice-role sequence:

```
guide
guide
storyteller
storyteller
guide
guide
```

4. exactly two practice questions,
5. every page has valid `imageID`,
6. every page has nonempty `imageDescription`,
7. six distinct page image IDs,
8. no duplicate resolved image bytes inside the same lesson,
9. existing 300-second total idea timing,
10. existing manifest/source/artwork/package limits.

Runtime import/storage remains tolerant of existing valid 2–40-page format-2 content.

Update at minimum:

- `LifeIsLearned/Models/LessonPackage.swift`
- `LifeIsLearned/Views/ReaderView.swift`
- `LifeIsLearned/Views/NarrationText.swift`
- `LifeIsLearned/Views/TakeawayCard.swift`
- `Tools/validate_package.py`
- `Tools/test_authoring.py`
- `CONTENT_AUTHORING.md`
- focused native/UI tests

Do not silently modify existing authored package files as part of this engineering patch.

9. Reader labels

Recommended stage language:

Screen	Reader label
Hook	Hook
Explanation	Explanation
Story A	Story · 1 of 2
Story B	Story · 2 of 2
Practical Application	Practical application
Takeaway	An idea to keep

Voice behavior remains driven by `NarrationRole`, not by visual labels.

10. Compatibility requirements

Must preserve:

- currently installed books,
- legacy page counts,
- current progress,
- collected Idea Cards,
- favorites,
- archived revisions,
- practice flow,
- Handoff 004 completion/continue behavior,
- manual Files import,
- formatVersion 2 whole-book packages,
- 1–12 ideas per new collection,
- Catalog 001 stable `book.id` values,
- current revision semantics.

Do not auto-rewrite old lesson content on app launch.

Do not reset progress because a newer authoring standard exists.

11. Tests and evidence

Add or update deterministic tests for:

Model / compatibility

- `.application` decodes and validates.
- legacy packages without `.application` still decode.
- stored legacy page counts remain readable.

Authoring gate

- canonical six-page sequence passes.
- five/seven pages fail authoring gate.
- wrong semantic order fails.
- wrong role order fails.
- missing image fails.
- duplicate image ID inside an idea fails.
- duplicate image bytes inside an idea fail.
- exactly two questions remain required.
- 300 seconds passes; >300 seconds fails.

Reader

- canonical six-stage lesson gets semantic indicator.
- legacy lesson gets old capsule indicator.
- Story A half-fills the wide Story segment.
- Story B fully fills it.
- Practical Application label is correct.
- Takeaway artwork renders.

Narration follow

- long Story text triggers narration-follow.
- Hook/Explanation/Application/Takeaway narration does not auto-drift at normal size.
- user drag prevents auto-scroll fighting.
- Reduce Motion uses nonanimated following.
- VoiceOver disables automatic follow behavior.

Regression

Run the existing relevant native/unit tests and Handoff 004 UI journeys.

A physical-device premium-voice run is not required solely to approve this UI/authoring-contract patch unless narrated script content changes. The first rebuilt six-screen book still receives normal release-quality timing verification.

12. Evidence report

Create:

Evidence/Handoff004_6/

Include:

- implementation commit SHA,
- exact baseline SHA,
- changed files,
- authoring validator/test output,
- native test output,
- relevant UI test output,
- screenshots of Hook, Story A, Story B, Practical Application, Takeaway-with-art, and a long Story auto-follow state,
- confirmation that legacy lessons retain fallback behavior.

Push the implementation/evidence commit and stop for reviewer approval.

Do not merge automatically.

13. Out of scope for 004.6

Do not add:

- remote catalog transport,
- GitHub Releases distribution,
- scanner/request workflow,
- Dive Deeper,
- bulk download management,
- cloud sync,
- book offload/storage reclamation,
- changes to Catalog 001 identities/queue,
- automatic migration/re-authoring of existing books.

The first content rebuild using this standard happens after reviewer approval.

14. Pinned Handoff 005 requirement — Book Offload

This is recorded now but should **not** be implemented inside 004.6.

User-facing behavior

Users must be able to **offload a book** from device storage without erasing what they learned.

Offload is not Delete.

When a book is offloaded, these must remain:

- all earned Idea Cards,
- favorite state,
- lesson/progress history,
- lightweight started/completed-book history,
- completion date when known,
- stable `book.id`,
- title/author snapshot sufficient for History UI,
- completed/practiced idea counts,
- progress identities needed to restore state after re-download.

These may be removed:

- lesson-package prose,
- full source definitions,
- package artwork,
- cover bytes,
- other heavy distribution payload.

Reinstall/redownload behavior

When the same stable book is re-imported/downloaded:

- earned cards do not duplicate,
- compatible progress reconnects by stable book/idea/revision identity,
- completion history remains,
- newer revisions use existing update semantics.

History UI

A completed offloaded book still appears in a lightweight **History / Completed Books** representation.

Idea Cards from an offloaded book remain browsable.

If the user requests the full lesson behind an Idea Card while its package is offloaded, offer to re-download/re-import rather than pretending full lesson content is local.

Architecture warning

The current storage design preserves historical `collections.json` snapshots, and `IdeaCardStorage.archivedSource(for:)` can retrieve full lesson packages from them.

Therefore merely removing a book from the active `CollectionCatalog` may **not actually reclaim storage** because old snapshots can still retain package/art bytes.

Handoff 005 must solve real reclamation deliberately by separating durable learner state/history/cards from removable package payload/artwork and adding safe garbage-collection/reference rules.

Do not ship an Offload button until tests prove storage is genuinely reclaimed while cards, progress, favorites, and history survive.

This aligns naturally with Handoff 005's planned remote download/redownload and storage-reclamation work.

15. Reviewer acceptance summary

Handoff 004.6 is accepted when:

- `.application` exists without breaking old packages,
- the six-page authoring contract is enforced at authoring gate,
- `Hook | Explanation | Story | Prac. App. | Takeaway` progress works,
- Story spans two physical pages and fills progressively,
- Story is the intended overflow/auto-follow stage,
- every new authored screen requires meaningful artwork,
- Takeaway shows artwork,
- old books work without forced migration,
- existing Handoff 004 learner state/cards remain intact,
- evidence/tests pass,
- offload is pinned for 005 but not mixed into this patch.

After approval, rebuild **one existing book** as the reference six-screen package before resuming batch production.